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System on Integrated Chips (SoIC™) for 3D Heterogeneous Integration

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Abstract—**A brand new 3D integrated circuit (3DIC) solution, System on Integrated Chips (SoIC™), has been successfully developed to integrate active and passive chips into a new integrated SoC system to meet ever-increasing market demands on higher computing efficiency, wilder data bandwidth, higher functionality packaging density, lower communication latency, and lower energy consumption per bit data. 3D packaging is challenging and requires overcoming three major challenges – thermal, power delivery, and yield. The SoIC, as industry-first 3D logic-on-logic and memory-on-logic chiplet stacking technology platform, enables the heterogeneous integration (HI) of known good dies (KGDs) with different chip sizes, functionalities and wafer node technologies, all to be integrated in a single, compact new system chip. From external appearance, SoIC looks like a general SoC chip with multiple pre-designed heterogeneous functional chips embedded. As SoIC is fabricated using "front-end" process, it can be holistically integrated into variant "back-end" advanced packaging technology platforms such as flip chip, integrated fan-out (aka InFO), 3DIC, and 2.5D with Si interposer (e.g. CoWoS™) [1-2] to provide a miniaturized and highly integrated HI SiP for the future HPC, AI, 5G, and edge computing applications.**

With the innovative bonding scheme, SoIC enables the strong bonding pitch scalability for chip I/O to realize a high density die-to-die interconnects. The bond pitch starts from sub-10 μm rule. Short die-to-die connection of SoIC has the merits of smaller form-factor, higher bandwidth, better power integrity (PI), signal integrity (SI), and lower power consumption comparing to the current industry state-of-the-art packaging solutions. In this paper, we demonstrated for the first time an integration of SoIC chip into InFO_PoP without increasing its form-factor. The SoIC was made on a logic-on-logic stacking to validate the design rules, process maturity, and reliability.

Keywords—3DIC; SoIC; PI/SI; 5G/AI; InFO_PoP; data bandwidth; sub-10um IO

I. INTRODUCTION

We have previously developed 3DIC TSV technology to directly stack logic on logic, and wide I/O memory on top of an application processor on an organic substrate for low power consumption, high data bandwidth, low latency, and improved PI/SI [3-5]. This was an industry-first effort led by leading foundry involving logic chips and/or memory chip stacking using TSV technology and fine pitch Cu pillar flip chip assembly technology ready prior 2012. Recent years, 3DIC integration technology has evolved from substrate package level to wafer level system integration (WLSI). This WLSI leads the semiconductor industry into a new era of system scaling beyond Moore's Law. High density integrated fan-out (aka InFO_PoP) in smartphone and high performance silicon interposer 3D chip stacking (aka CoWoS) in HPC and AI have been widely adopted in market segments for years [6-8]. The challenges of Moore's Law of advanced node SoC become more stringent and some critical functions in the SoC become more difficult to scale with the core devices. Those issues can be mitigated by SoIC technology with partition and chiplets approach [9-12].

II. SoIC™ 3D INTEGRATION

SoIC technology enables the system scaling with better merits of cost and performances by implementing SoC partition and re-integrating through the stacking of I/O logic chip and core circuits chip. SoIC also offers the design and integration flexibility for mix and match of heterogeneous chips in different technology nodes, materials, functionalities, and chip sizes to create a true heterogeneous 3DIC. The SoIC envelope to expanding its chip size opens new opportunities for highly integrated system as shown in Fig.1. With the closest proximity between two chips through direct chip-to-chip interconnect, the SoIC creates a superior power integrity, signal integrity and a much lower communication latency with more than 20Tbps memory bandwidth to support the future HPC, AI, 5G, and edge computing applications [13]. The ultra-fine pitch of SoIC bond can be scalable down to sub-10 micron to enable chip functional block level ultra-high density interconnects that cannot be achieved by the typical 3D IC TSV with flip chip bond technology [7-8, 14, 15].

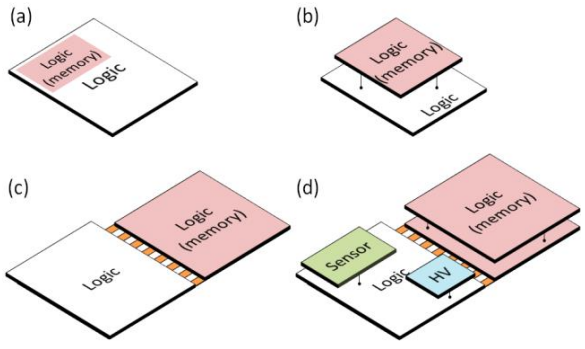


Figure 1. (a) SoC before chip partition; (b), (c), (d) Variant partitioned chiplets and re-integrated schemes enabled by SoIC

Leveraging leading foundry wafer front-end process and WLSI infrastructure, the SoIC manages the challenging thin

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die handling and bonding in advance and leverages well-established WLSI package integration to achieve high integration yield and reliability beyond the typical 3DIC stacking approach as shown in Fig. 2.

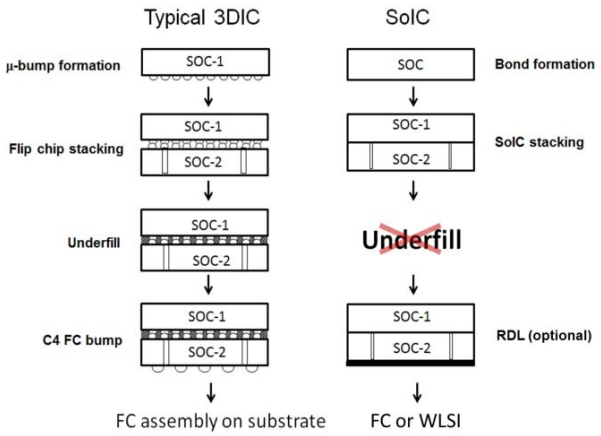


Figure 2. Comparison of integration flow between typical 3DIC package and SoIC package

In 3D stacking, chiplets are interconnected in face-to-face (F2F), face-to-back (F2B), and a combination of both. SoIC employs KGDs for 3D heterogeneous integration. Chiplets with different sizes, different functions/technology nodes can be integrated in 2D side-by-side manner for mix and match design flexibility and in 3D stacking manner for integration extendibility. Fig. 3a-3b demonstrates the design flexibility and integration capability of SoIC with different chiplets sizes. More than that, Fig. 3c demonstrates the extendibility of SoIC in 3-tiers of logic-on-logic stacking, which can be theoretically extended to more tiers depending on design needs. Fig. 3d shows the typical 3DIC flip chip stacking with μbumps, which leads to an obvious higher profile than SoIC does.

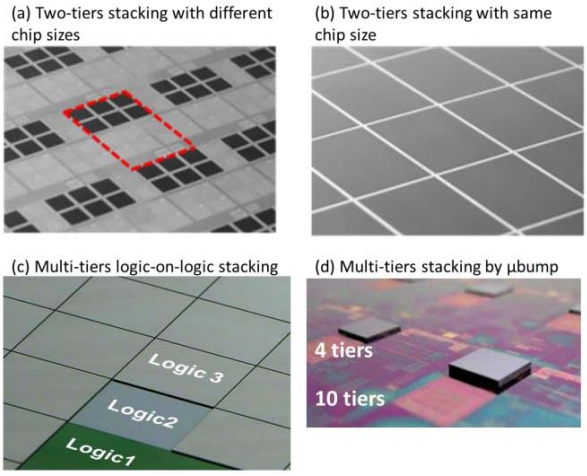


Figure 3. SoIC bond/stacking with (a) different small chiplets (b) the same large chiplets, (c) 3-tiers logic-on-logic integration, (d) typical 3DIC stacking with μbump bond

III. SoIC™ PACKAGE INTEGRATION

SoIC integrates both homogeneous and heterogeneous chiplets into a single SoC-like chip with a smaller footprint and thinner profile, which can be holistically integrated into any advanced package platforms such as flip chip, and WLSI (aka CoWoS [16] and InFO [17-18]). From external appearance, the SoIC is just like a general SoC chip yet embedded with desired and heterogeneously integrated functionalities (see Fig. 4)

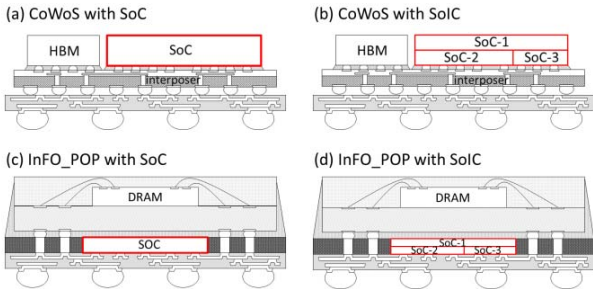


Figure 4. (a) CoWoS with SoC, (b) CoWoS with SoIC integrating partitioned SoCs (c) InFO_PoP with SoC, (d) InFO_PoP with SoIC integrating partitioned SoCs

IV. REALIZATION OF InFO_PoP WITH SoIC™

In this paper, we have successfully demonstrated the industry-first SoIC with logic-on-logic stacking integration in InFO_PoP platform. Compared to the typical 3DIC PoP, the SoIC-embedded InFO_PoP offers higher interconnect I/O bonding density, lower power consumption and thinner package profile. Fig. 5 shows the system scaling of bump/bond density can be significantly boosted by SoIC technology to unleash the limitation of flip chip bonding density beyond 10K/mm^2. The limitation of bump/bond density on flip chip bond lies in two bottlenecks. One is the fine bump pitch limitation due to the inadequacy of bumping tools and materials. The other is the reliability issue due to the so-called chip-package-interaction (CPI) in fine bump pitch flip chip assembly.

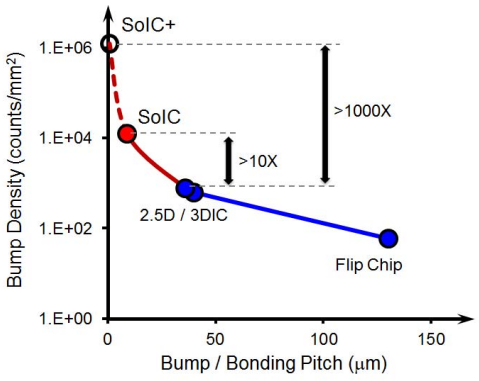


Figure 5. Leapfrog boost of SoIC on system bump/bond density

With foundry front-end process control, the integration of SoIC achieves high bonding yield. Fig. 6 shows the measured

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high bonding yield upscale to several millions SoIC bonding I/O counts.

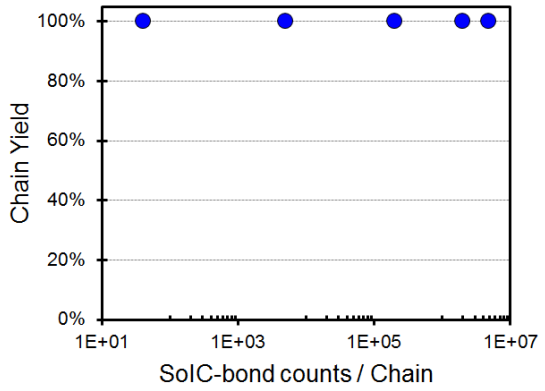


Figure 6. SoIC achieves high bonding yield upscale to several millions I/O counts/ chain

V. SYSTEM PERFORMANCE BENCHMARK

System performances such as interconnect bump/bond density, electrical performances (aka PI/SI), and thermal performance are critical to meet mobile and HPC system designer's demands on high computing efficiency, data bandwidth, low latency, low energy consumption per bit data operation. In the next session, the comparisons between SoIC bond and typical 3DIC stacking with µbump flip chip bond on aforementioned system performances are presented to realize the enablement of SoIC technology beyond the 3DIC stacking.

A. Interconnect Bump/ Bond Density

Fig. 7 shows the comparisons of interconnect bump/bond pitch and density between typical 3DIC and SoIC. According to our analysis, SoIC bond outperforms µbump of 3DIC at least by 4X smaller in bond pitch, and minimum by 16X higher in bond density, respectively in normalized scale. Higher interconnect bonding density allows chip designer and system designer to have the flexibility and extendibility on designing a new chip and new system device adopting the most state of the art integration technology for cost as well as performances.

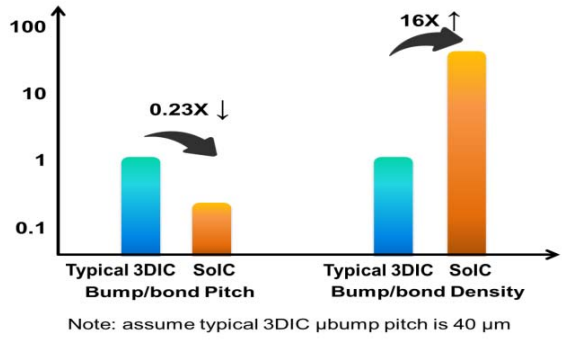


Figure 7. Bump/Bond comparison between SoIC bond and µbump of typical 3DIC

B. Electrical Performance

In adopting a new interconnect technology, the electrical resistance and its stability are critical to package designer. Fig. 8 compares the excellent cumulative percentage variations of electrical resistance of SoIC bond, and TSV/TDV with foundry mature Cu dual damascene via. The result reflects the compatibility and process readiness/robustness of SoIC bond, TSV, and TDV for customer designs and applications.

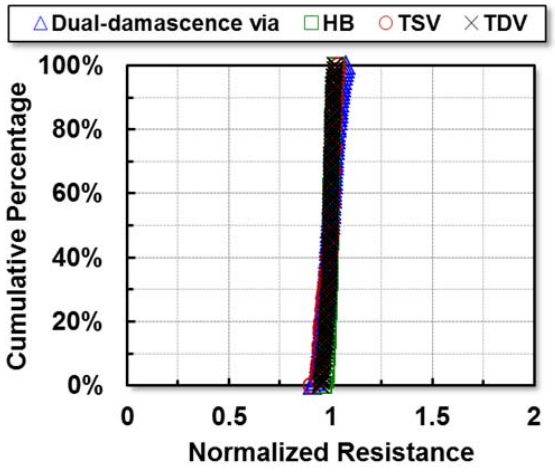


Figure 8. Normalized electrical resistance comparison between Cu dual damascene and SoIC bond/TSV/TDV

Low insertion loss is essential to enable the interconnect performance in high speed, high frequency data transmission between two SoC chips, particularly in 5G communication applications. Fig. 9 compares the insertion loss between SoIC bond and 2D side-by-side flip chip bond at the frequency less than 30 GHz. SoIC bond exhibits almost no insertion loss throughout the 30 GHz spectrum, while flip chip bond extends the insertion loss as frequency increases.

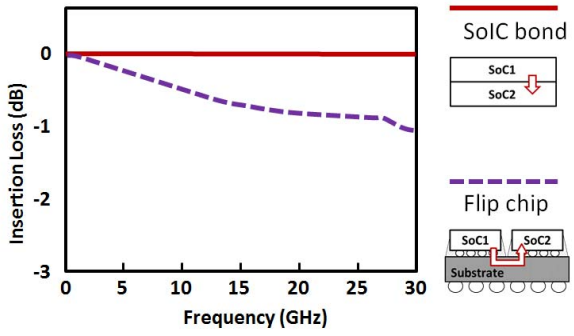


Figure 9. Insertion loss of SoIC and flip chip spans 30GHz frequency spectrum

Aside from the insertion loss, the interconnect wiring parasitics of R, L, and C for fine pitch, high density bonding is very important to ensure the high quality of signal integrity and power integrity with low RC delay and low IR drop. Fig. 10 compares the wiring parasitics between the µbump of typical

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3DIC stacking and SoIC bond. According to our analysis, SoIC bond outperforms µbump of 3DIC, at 2GHz, by 12.5X in R, by 100X in L, and by 12.5X in C, respectively in normalized scale. As a result, SoIC bond leads µbump bond by 156X in RC delay, and by 12.5X in IR drop.

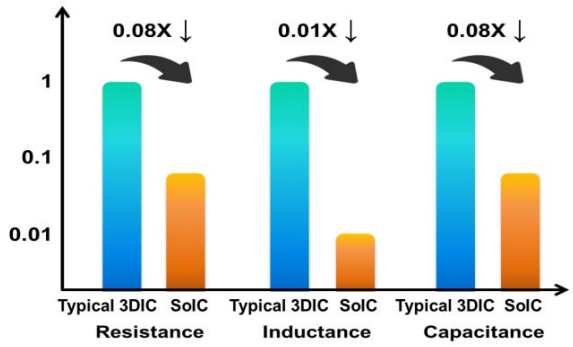


Figure 10. R, L, C wiring parasitics comparison between SoIC bond and µbump of typical 3DIC

C. *Thermal/Power Performance*

Thermal management becomes increasingly important in heterogeneous system integration, particularly for 3D chip stacking, due to a high thermal power density challenge. Exceeding the allowable working temperature, the chip may fail to function normally and cause severe leakage current issue. Thus engineers from advanced packaging community have exhausted means to resolve the thermal issues from 3D chip stacking. Compared to the typical 3DIC stacking, SoIC by its nature offers higher metal routing density and enables thermal flow in both upward and downward directions, and thus can dissipate the thermal flux more effectively (see Fig. 11). According to our analysis, SoIC bond outperforms µbump of 3DIC by 11X in energy consumption/ bit data, and by 1.25X~2X in thermal resistance, respectively in normalized scale. The significant reduction in the amount of energy required to move data around is very meaningful. At a fixed power envelope, the less energy spent per bit data, the more bits can be transferred. Alternatively, saved energy can be spent on other resource uses.

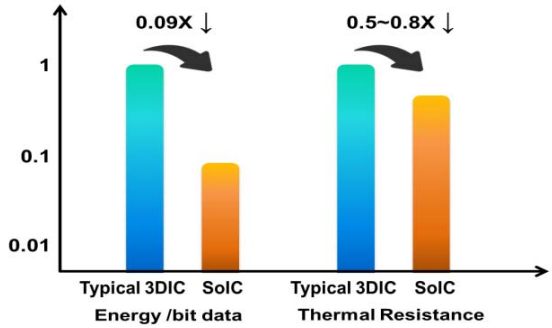


Figure 11. Comparison of thermal performance between SoIC bond and µbump of typical 3DIC

VI. RELIABILITY

To ensure the structural integrity of fine pitch, high density interconnects on SoIC bond, TSV, and TDV (through dielectric via), the chip level reliability of InFo_PoP with SoIC on electro-migration (EM), stress-migration (SM), and break down voltage (Vbd) were conducted. The test results are plotted in Fig. 12, Fig. 13, and Fig. 14, respectively. According to test results, TSV, TDV, and SoIC bond demonstrate robust reliability performances to pass TSMC internal requirements on EM, SM, and Vbd throughout the test conditions. Fig. 15 further shows SoIC to pass the PCT/168hrs, HTS/1000x, TCC/1000x, and uHAST/168hrs.

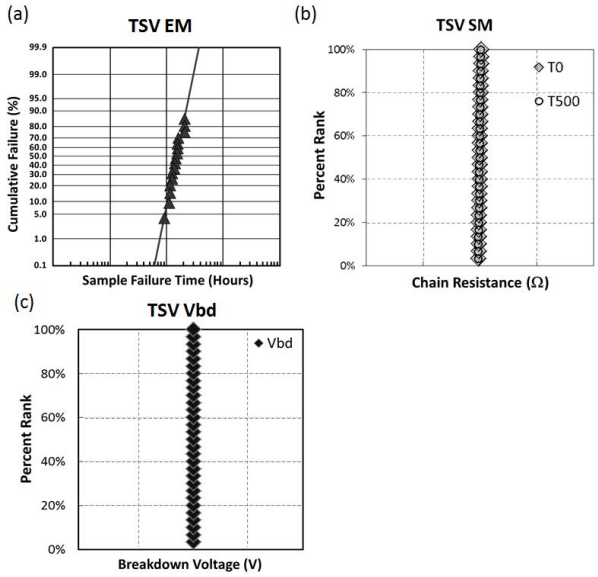


Figure 12. TSV reliability on (a) EM, (b) SM, and (c) Vbd

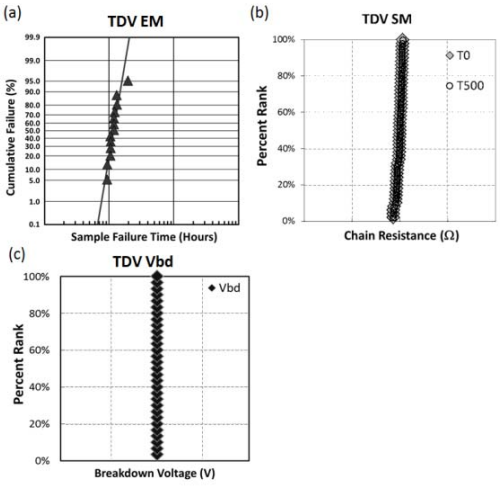


Figure 13. TDV reliability on (a) EM, (b) SM, and (c) Vbd

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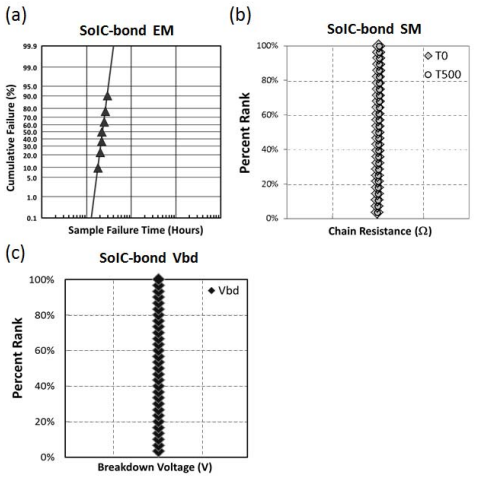


Figure 14. SoIC bond reliability on (a) EM, (b) SM, (c) Vbd

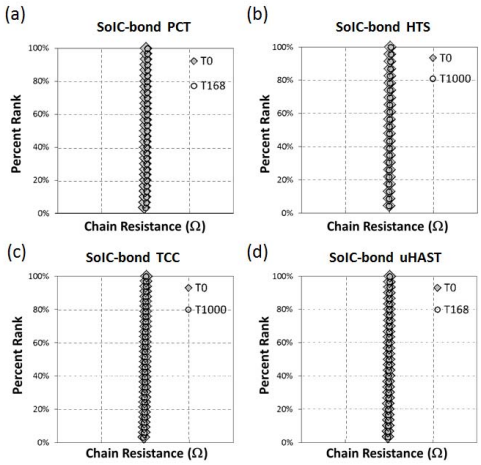


Figure 15. SoIC bond reliability on (a) PCT, (b) HTS, (c) TCC, and (d) uHAST

At package level, Table 1 tabulates various reliability tests being carried out on SoIC-embedded InFO_PoP. All test samples have passed the standard JEDEC test conditions and tests with extended test cycles and time. No failures were found after TC-B (1000x) and uHAST (192 hours).

Table 1. SoIC in InFO PACKAGE LEVEL RELIABILITY

Reliability	Test Conditions	Duration/ Cycle	Result
Quick Torture	MR (multi-reflow @ 260°C)	10x	Pass
	MR10x + TCC (-65~150°C)	200x	Pass
	MR3x + uHAST	96 hours	Pass
CLR	MSL	Level 3	Pass
	TC-B (-55~125°C)	700x	Pass
		1000x	Pass
	uHAST (130°C, 85% RH, 33.3-psi VP)	96 hours	Pass
		192hours	Pass

VII. CHALLENGES

There are challenges on design and process of SoIC integration. These challenges are very critical yet manageable for us by leveraging the well-established WLSI infrastructure in know-hows, materials, tools, and design rules. Many studies were conducted to characterize the impacts of TSV along with the size of keep out zone on IC transistors from thermal and mechanical aspects. In advanced node SoC, the silicon asset is precious for function blocks and IP designs. Large size TSV will consume more silicon area and increase the fabrication cost of advanced node SoC. Thus design rules with finer feature size of TSV are critical challenges to be addressed.

VIII. CONCLUSIONS

We have successfully developed and demonstrated the industry-first logic-on-logic and memory-to-logic 3D stacking using the SoIC technology. In comparison to the typical 3DIC stacking, SoIC offers higher I/O bonding density beyond the limitation of μbump flip chip bond, and enables lower energy consumption/ bit data, lower electrical parasitics in RC delay/IR drop, lower thermal resistance to unleash the boundary of IC designer on future heterogeneous integrations in 5G, AI, mobile, and HPC applications. Furthermore SoIC technology adopts the foundry front-end 3D process and leverages the well-established infrastructure of 3D WLSI in materials, tools, processes, design rules to create a holistic 3D WLSI family via CoWoS, InFO, and flip chip to shorten the development cycle time with high yield and competitive cost. To this end, we have demonstrated in this paper a SoIC-embedded InFO_PoP with verified system performances, robust reliability, and high stacking yield. SoIC becomes one of the most important WLSI technologies in future 5G and AI-driven applications for years to come.

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